

SECTION 15094

PIPING SUPPORTS AND ANCHORS

PART 1 - GENERAL

1.1 WORK INCLUDED

- A. This Section specifies the requirements to design, furnish and install pipe and equipment hangers and supports which includes; hangers, spring hangers, guides, anchors, supports, slides, and bracing for pipe and piping system components.

1.2 RELATED WORK

- A. This Section shall be used in conjunction with the following other specifications and related Contract Documents to establish the total requirements for piping supports and anchors:
 - 1. Section 15023 – Codes and Standards.
 - 2. Section 15048 – Mechanical Painting.
 - 3. Section 07841 – THROUGH-PENETRATION FIRESTOP SYSTEMS.
 - 4. Section 15090 – PVC and CPVC Piping Systems.
 - 5. Section 15091 – PVDF Piping Systems.
 - 6. Section 15092 – PFA Piping Systems.
 - 7. Section 15101 – Carbon Steel Piping.
 - 8. Section 15200 – Vibration and Noise Control.
 - 9. Section 15372 – High Purity Stainless Steel Tubing Systems.
 - 10. Section 15373 – General Purity Stainless Steel Tubing Systems.
 - 11. Section 15375 – Copper Tubing Systems.
 - 12. Section 230500 – Common Work Results for HVAC.
 - 13. Section 230700 – Equipment, Vessels, Piping, and Duct Insulation.

1.3 DESIGN CRITERIA

- A. The piping supports and anchors shall be in accordance with the requirements of the following as applicable:
 - 1. Manufacturers Standardization Society MSS-SP-58, Pipe Hangers and Supports - Material, Design and Manufacture.
 - 2. Manufacturers Standardization Society MSS-SP-69, Pipe Hangers and Supports - Selection and application.
 - 3. Manufacturers Standardization Society MSS-SP-89, Pipe Hangers and Supports - Fabrication and Installation Practices.
 - 4. ASME B31.9 – Building Services Piping.
 - 5. AWS – American Welding Society D1.1
 - 6. PCI Design Handbook for Concrete Inserts Design.
 - 7. NFPA 13 - Standard for the Installation of Sprinkler Systems.
 - 8. NFPA 14 - Standard for the Installation of Standpipe and Hose Systems.
 - 9. Details and Drawings.
 - 10. All local codes and amendments.
- B. Supports, Anchors, and Restraints:
 - 1. When supports and/or anchors for equipment, valves, conduit, piping and

- appurtenances are not shown on the Drawings; the Contractor shall be responsible for their design and installation, with approval from the Engineer. Supports, anchors, guides, and locations of hangers for chilled water, hot water, IW, and UPW shall be designed by the Engineer.
2. Connections to structural framing shall not introduce twisting, torsion, or lateral bending in the framing members. Design of supplementary support steel shall be by the Engineer.
 3. Where indicated on drawings, "Contractor Engineered Support Systems" shall be designed and detailed with required submittals bearing the seal of a Professional Engineer registered in the state of Idaho, regularly engaged in this type of work.

1.4 SUBMITTALS

- A. Provide the following prior to installation with the Submittals:
 1. Manufacturers descriptive literature, including manufacturer's name, model number, description of materials, and intended service, for all proposed items including product data and equipment supports including rods, pipe shields, clamps, springs, inserts, etc.
 2. Submit fabrication drawings or manufacturer's literature and proposed location of trapeze hangers, pipe anchorage, and auxiliary bracing as required.
 3. Details for the piping attachment methods to rack type frames and trapeze supports (i.e. slides, guides, rollers, u-bolts, etc.).
 4. When special installation or fabrication conditions arise, submit shop drawings and other descriptive data detailing proposed methods of installing hangers, supports, and anchors for review by Owner's Representative prior to installation.
 4. Submit shop drawings of reinforcing structural steel for anchor points and any other locations requiring reinforcement.
 5. Shop drawings of suspended piping, indicating point loads along with applicable pipe support details keyed to layouts.
 6. Shop drawings for pipe support frames, equipment supports, guides, expansion joints and anchors along with engineer's calculations keyed to the layouts drawings.

1.5 QUALITY ASSURANCE

- A. Conform to Section 15023 – Codes and Standards for all International, State and local codes.
- B. Coordination: Coordinate with the installing contractor for mechanical insulation to establish the insulation material and thickness that will be furnished for each insulated system.
- C. Qualify welding processes and welding operators according to AWS D1.1 "Structural Welding Code – Steel".
- D. Certify that each welder has satisfactorily passed AWS qualification tests for welding processes involved and, if pertinent, has undergone recertification.

1.6 DEFINITIONS (SPECIFIC TO THIS SECTION)

- A. Hot Pipes: Pipe systems with operating temperature greater than 100 degrees F.
- B. Cold Pipes: Pipe systems with operating temperature less than or equal to 100 degrees F.

PART 2 - PRODUCTS

2.1 ACCEPTABLE MANUFACTURERS (or Approved Equal)

- A. Pipe Hangers and Supports:
 - 1. Anvil
 - 2. TOLCO
 - 3. PHD
 - 4. Superstrut
 - 5. Pipe Shields
 - 6. B Line.
 - 7. Unistrut.
 - 8. Fronek Company
 - 9. Piping Technology and Products
 - 10. Bergen – Paterson
 - 11. Approved Equal
- B. Pipe Hanger Isolation Shields:
 - 1. Pipe Shields Incorporated.
 - 2. Insul-Shield.
 - 3. Rilco
 - 4. Bergen – Paterson
 - 5. Valued Engineered Products
 - 6. Approved Equal
- C. Metal Framing Systems:
 - 1. Unistrut.
 - 2. Superstrut.
 - 3. Kin-Line.
 - 4. Hilti
 - 5. Power Strut
 - 7. B-Line.
 - 8. Approved Equal.
- D. Alignment Guides:
 - 1. Anvil
 - 2. Pipe Shields Inc.
 - 3. Rilco
 - 4. Bergen - Paterson
 - 5. Approved equal
- E. Pipe Cushions/Clamps:
 - 1. ZSI
 - a. Cush-a-strip

- b. Cush-a-clamp
 - 2. Coper B-Line
 - a. Iso-Pipe Tape
 - b. Armafix
- F. Sleeves:
 - 1. Thunderline Corp.
 - 2. Pipe Shields Inc.
 - 3. Calpico
 - 4. Approved Equal

2.2 MATERIAL AND FINISHES

- A. All Slotted Channel Framing shall be 316 stainless steel or powder coated in the "FAB" (area above raised floor). All hangers, rods, clamps, protective shields, metal framing supports components, hardware, and hanger accessories shall be 316 stainless steel. Material shall be approved by Owner Prior to purchase.
- B. All Slotted Channel Framing shall be Gold Guard (Perma-Gold, Yellow dichromate Conversion Coat) in the "SUB-FAB", "SUPPLY PLENUM", electrical rooms, mechanical rooms, office ceiling space, and other non-corrosive environments. All hangers, rods, clamps, protective shields, metal framing support components, hardware, and hanger accessories shall be electro-galvanized (electro-plated Zinc). Material shall be approved by Owner prior to purchase.
- C. All Slotted Channel Framing shall be fiberglass in corrosive environments such as chemical rooms, chemical containment areas, chemical dosing rooms, for process waste-waters, process and mechanical HVAC wet treatment and cooling towers. Alternates for chemical rooms to be approved by owner.
- D. All slotted channel framing shall be galvanized steel in outdoor non-corrosive environments.
- E. Refer to Section 15048 – Mechanical Painting for primer and coating requirements.

2.3 GENERAL

- A. Provide pipe hangers, supports, and accessories which.
 - 1. Permit vertical adjustment after installation of piping.
 - 2. Are designed for support of piping and contents under all conditions of operation.
 - 3. Will not crush, indent or otherwise damage pipe, pipe insulation, or jacketing.
 - 4. Utilize all similar support components by same manufacturer.
 - 5. Provide complete hanger assemblies and support assemblies, including clamps, clevises, structural attachments, rods and pipe rolls, constructed for compatibility with items supported.
- B. Provide oversized hangers for all insulated piping to accommodate insulation thickness specified in Specification Section 230700 – Equipment, Vessels, Piping, and Duct Insulation.

- C. Provide Insulation protectors at support points for all insulatedunjacketed piping.
- D. Do not support piping with wire either temporarily or permanently.
- E. Do not support piping over (8") eight inches in diameter with slotted channels. For larger line sizes, including multiple lines on the same strut-type channels require Engineer of Record review and approval, as well as approval by Owner.
- F. The maximum deflection of a channel member supporting at midpoint of span, shall not be greater than 1/4 inch.
- G Use the following data, the load limits published by the hanger components manufacturer or Table 1 for minimum rod size requirements. Contractor to adhere to whichever is most stringent.

<u>Max. Total Load Per Rod</u> ⁽¹⁾	<u>Min. Hanger Rod</u>
610 lb.	3/8 in.
1130 lb.	1/2 in.
1810 lb.	5/8 in.
2710 lb.	3/4 in.
3770 lb.	7/8 in.
4960 lb.	1 in.
8000 lb.	1-1/4 in.
11,630 lb.	1-1/2 in.
15,700 lb.	1-3/4 in.
20,700 lb.	2 in.

⁽¹⁾ including trapeze, attachments, pipe weight, fluid in pipe (water), insulation

2.4 PIPE HANGERS AND SUPPORTS

- A. Pipe hanger and support components are listed below. Where more appropriate items are required to accommodate the pipe loading, displacements, or other conditions, submit these items to the Engineer.
- B. Individual Hangers for Cold Pipe up to 30 Inch Pipe Size: adjustable, clevis type; Anvil Fig. 260 or equivalent, oversized where required to fit on outside of insulation; SS pipe use SS materials.
- C. Individual Hangers for Hot Pipe: insulated 2 bolt hanger; Pipe Shields, Model D1000 through D6300 or equivalent.
- D. Trapeze Hanger Type Supports: Construction of field fabricated back to back channels shall be compatible with Anvil channel assembly Fig 45, 50, or equivalent.
- E. Vertical Support- Uninsulated Piping: steel riser clamps. Anvil Fig 261 with steel members, and Fig 40 with rods or equivalent. Minimum of two shear lugs sized to accommodate the applicable load, welded to pipe 180 degrees apart are required for all riser clamps. The lug material to be compatible with pipe materials.

- F. Vertical Supports- Insulated Piping: insulated steel riser clamp. Pipe Shields, Model E1000 through E2330, or equivalent
- G. Floor Support for Pipe Systems to 4 Inches and All Cold Pipe Sizes: cast iron adjustable pipe saddle, locknut nipple, floor flange, and concrete pier or steel support; Anvil Figure 264, black only, or equivalent
- H. Floor Support for Hot Pipe Sizes 6 Inches and Over: adjustable cast iron roll and stand, steel screws, and concrete pier or steel support.
- I. Copper Pipe Support: steel ring, adjustable, copper plated; Anvil Figure CT-99, CT-65, or plastic-coated Anvil Figure CT-99C, or equivalent.
- J. Nonmetallic Piping Supports: Special protection shields and neoprene pads shall be provided as recommended by the pipe manufacturer.
- K. Sliding Type Supports: PTFE slide plates consisting of steel tees, plates and PTFE sliding surfaces for free movement of piping due to expansion and contraction; Anvil figure 247, Pipe Shields Model B 1000 through B2300 for insulated piping, or equivalent.
- L. Slotted Channel Framing: Provide pre-galvanized channel unless noted otherwise in section 2.2 of this specification.
- M. Pipe Vibration Spring Isolators: Per requirements of Section 15200 - Vibration and Noise Control.
- N. Steel Hanger Rods: Threaded both ends, or continuous threaded. Anvil Figure 140 or 146, or equivalent. Electrogalvanized rods for corrosion resistant areas or areas exposed to moisture unless otherwise stated in section 2.2 of this specification.
- O. Anchors: Pipe Shields Model C4000 through C4300 for insulated Ferrous piping, or equivalent.
- P. Pipe Attachments:
 - a) Clevis Hangers: steel, oversized where required to fit on outside of insulation, Anvil Fig. 260.
 - b) Roller Hanger:
 - i. Clevis type with swivel, yoke, cast iron roller, Anvil Fig. 181.
 - ii. Two rod type, cast iron adjustable sockets and roller Anvil Fig. 171 with hanger rods and nuts.
 - c) Ring Hanger: Split type, Anvil Fig. 104.
 - d) Double Bolt Hanger: Anvil Fig. 295, 295H.
 - e) Adjustable Pipe Ring: Anvil Fig. 97, 97c (plastic coated).
 - f) Adjustable Swivel ring: Anvil Fig. 69.
 - g) Riser Clamp: Anvil Fig. 261.
 - h) U-Bolts: Anvil Fig. 137.
 - i) Band Hanger: Anvil Fig. 69.
 - j) Roller Chair: Anvil Fig. 175, 271, 274, 275, 277.
 - k) Wall Hook Plate: Shop or field fabricated from bar stock and "J" hooks to suit application.

- l) Roller with Saddle: Anvil hanger standards 75SD, 76SD.
- m) Floor Supports:
 - i. Pipe support carbon steel, Anvil Fig. 258, 259, 264; include support riser pipe welded to baseplate.
 - ii. Base elbow support stanchion, carbon steel pipe welded to square steel baseplate.
- n) Plate Washers: Anvil Fig. 60

- Q. Pre-engineered Spring Hangers (where indicated): Pre-engineered, short spring type with load indicator, load scale and nameplate, Anvil Fig. B-268, 82, 98.

- R. Hanger Rods: Hot rolled steel, ASTM A36.

- S. Turnbuckles: Anvil Fig. 230.

- T. Rod Couplings: Anvil Fig. 136, 136R.

- U. Inserts:
 - a) Universal Type: Anvil Fig. 282, 281.
 - b) Continuous Type: slotted steel.
 - c) Screw Type: Anvil Fig. 152.

- V. Expansion Fasteners: Self-drilling, steel shell and expander plug.

- W. Beam Clamps: Anvil Fig. 228, 229, 292; provide C-clamps (MSS types 19 and 23) only where no other means of attachment is possible. All C-clamps shall have restraining straps.

- X. Guides:
 - a) Cylinder or Spider Type: Anvil Fig. 256.
 - b) Slide Plate Assembly: Anvil Fig. 257.

- Y. Hollow Core Anchors: ITT Phillips TG series.

- Z. Precast Anchors: ITT Phillips WR series.

- AA. Channels for Plastic Pipe: 18-gauge steel, broken to 90 ° angle at midpoint.

- BB. Provide all required suitably constructed sway bracing, guy wires, couplings, and attachments as required to prevent excessive vibration and damage.

- CC. Provide miscellaneous structural members as required for finished installation of hangers and supports.

- DD. Provide 1/8-inch-thick rubber pads or equivalent to prevent contact of stainless steel and copper pipe with black carbon steel hangers and surfaces of carbon steel pipe.

- EE. Provide supports for nonmetallic pipe that are non-damaging to pipe surface.

- FF. Fit all "Unistrut" and all-thread or approved equal with plastic end per section 3.1.J. Trim all-

thread back to within 2 to 3 inches of hangers' threaded nut.

GG. Insulation Protectors:

- a) Cold Lines with Operating Temperatures of 100 ° F or Less: Rigid foam insulating saddle or section of cellular glass pipe insulation 1 foot long with galvanized steel pipe covering protection shield. (Anvil Fig. 167).
- b) Shield for Insulated Piping 2 Inches and Smaller: 18-gauge galvanized steel shield over insulation in 180 degree segments; Anvil Figure 167, or equivalent.
- c) Carbon Steel Piping with Operating Temperatures of 100°F or Higher:
 - a) Steel pipe covering protection saddles (Anvil Fig. 160 through 165) for pipes supported with clevis or roller type hangers.
 - b) Structural tees, 1 foot long, height minimum of 1 inch more than insulation thickness, for pipes supported on trapeze hangers.
3. Non-Ferrous or Galvanized Steel Piping with Operating Temperature of 100°F or Higher: 360° rigid foam insulating saddle 1 foot long (Anvil Fig. 160, 161, 162) or section of cellular glass pipe insulation 1 foot long with galvanized steel 360 ° pipe covering protection shield.
4. All insulation protectors shall be the same thickness as pipe insulation on pipe where such protectors are used.

HH. Insulated pipe supports shall be utilized on insulated pipe lines. Proper models to be selected by hanger load, hanger style, and permissible hanger span distances specified herein.

- a) Model designations are by Pipe Shields, Inc., Rilco, Valued Engineered Products, or Engineer of Record and Construction Manager approved equal. Equals must submit documentation demonstrating compliance with applicable MSS and ANSI standards.
- b) Models:
 - i. Insulated Pipe Supports:
 1. Pipe Supported on Rod, Clevis and J-Type Hangers:
 - a. 1/2 inch through 4 inch: Pipe Shields Inc. Model D2000, A2000, A4000 or A9000, Rilco Model HC2000 (Glycol Hot water), CC2000 (Chilled Water), Valued Engineered Products Inc. Pro-Shield.
 - b. 5 inch through 10 inch: Pipe Shields Inc. Model A4000, A9000 or D3100, Rilco Model HC2000/ CC2000, Valued Engineered Products Inc. Pro-Shield.
 - c. 12 inch through 16 inch: Pipe Shields Inc. Model D3100, D3000, Series, D6000 through D6100, Rilco Model HC9000/ CC2000, Valued Engineered Products Inc. Pro-Shield.
 - d. 18 inch: Pipe Shields Inc. Model D3100, D3000, D3300 Series, D6000 through D6300, Rilco Model HH3000/ CC2000, Valued Engineered Products Inc. Pro-Shield.
 - e. 20 inch through 30 inch: Pipe Shields Inc. Model D3300, Rilco Model HH3100, HH3200/ CH1100, 32 inch through 42 inch: Pipe Shields Inc. None Available. Rilco Model HH3100, HH3200, / CH1100.
 - c) Pipes supported on pipe rollers and/or flat surfaces:
 - i. Note: Line sizes greater than 4 inch will require additional load/wear distribution

plates. Load distribution plates will be form-rolled to fit the outside, bottom-third section of the insulated support metal vapor-barrier shield. Plates to be tack or spot welded to the insulation shield.

d). Pipe Supports on pipe rollers and/or flat surfaces (H = hot fluids / C= cold fluids / R= roller support / F= flat support)

1. 4 inch pipe and below: Pipe Shields Inc. Model (R&F) A-4000 Rilco Model HR-4000, HF2000/ CR-CF 4000, Valued Engineered Products Inc. Pro-Shield D.S., or Pro-Shield N.T./D.S.
2. 5 inch to 8 inch: Pipe Shields Inc. Model (R) A-6000, (F) A-7000, Rilco Model HR-8000, HF-6000/ CR-4100/ CF 4000, Valued Engineered Products Inc. Pro-Shield WDP
3. 10 inch to 16 inch: Pipe Shields Inc. Model (R) A-8000, A-8200; (F) A7000, Rilco Model HR-8000, HR-8200/ HF-7000/ CR-4100, CF4000, Valued Engineered Products Inc. Pro-Shield WDP.
4. 18 inch to 28 inch: Pipe Shields Inc. Model (R) A-8200, A-8400/ (F) A-7200, A-7400, Rilco Model HR-8200, HR-8400/ HF7200/ CR-4100, CF-4000.
5. 30 inch to 42 inch: Sized, load-rated; Anvil, Superstrut or PHD weld-on pipe covering protection saddles with insulation inserts.

e). Guide shoes:

1. 1/2 inch through 30 inch: Pipe Shields Inc. Model B3000 to B3300 Series and B4000 to B4300 Rilco Model HG3000 through HG4000 as required restricting lateral movement.
2. 32 inch through 42 inch: Pipe Shields Inc. Model B3000 to B3300 Series and B4000 to B4300, Rilco Model HG3100

f). Anchors:

1. Pipe Shield Inc. Models C4000 through C4300 or as per drawing, Rilco HA4000 through HA4310 or as per drawing.

g). Pipe risers:

1. On Steel pipe 2 inch through 22 inch (welding required) use Pipe Shields Inc. Models E1000 through E1300 or E2000 through E2300 or Rilco HV2000 through HV2300 series, as required to prevent lateral movement and to accommodate upward and/or downward movement per specification and manufactures recommendation.
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- 3). Install as a minimum, hard packed, load-rated 360 degree calcium silicate with a density of not less than 9 pcf and a compressive strength of not less than 125 psi at 10 percent compression type saddles with properly sized banded shields.
 - 4). Model selections will be based upon support, vertical, longitudinal and axial load ratings. Unless noted on drawings; Selected models shall be submitted to

Engineer of Record and Construction Manager for Engineering approved.

- 5). As a minimum, all pipe sizes 10 inch and larger will require 400 psi minimum compressive strength structural inserts for all insulated appliance applications.
- 6). Piping in cleanroom areas will require insulation inserts to be fully coated with mastic such as Childers CP-11 or equal to provide a dust-free unit.
- 7). Cleanroom (fire protected) applications require 0.020 PVC jacketing between the banded metal shields and the 360 degree inserts.
- 8). Insulated supports shall be pre-assembled with sufficient insulation extending beyond the support outer metal shield so as to allow the pipe insulator to make a standard insulation butt-joint. Pipe with fluid 30 degrees F to 55 degrees F shall be provided with high density load rated (minimum 4.5 safety factor) polyurethane insulated inserts.
- 9). Supports will have tri-laminated or matching all service jacketing, adhere between the insulation insert and thermal metal shield.
- 10). All insulated supports shall match the adjoining insulation thickness. Note: Model designations are by Pipe Shields Inc., Value Engineered Products Inc., Rilco, or Engineer of Record and Construction Manager approved equal.
- 11). The person installing the pipe shall install insulated pipe supports for proper hanger and uniform support loading.
- 12). Insulated support halves shall be positioned to support the load of the pipe with the cut halves in the horizontal position.
- 13). Insulated inserts that are cracked, crushed or broken will not be used.
- 14). All flat and/or roller applications require 360-degree metal vapor shields stainless steel banded to the pipe after installation. Note: When banding insulated supports to the pipe, exert a minimum clamping force on the pipe to assure that the assembly will move axially with the pipe and not slip relatively to the pipe when subjected to the design conditions.

2.5 CONCRETE EXPANSION ANCHOR BOLTS

- A. Hilti - Kwik II, Drop - ins, may be used for non-dynamic pipe support in accordance with the manufacturer's recommendation and the PCI Design handbook.
- B. Pipe support anchors for dynamic pipe support must be either an approved adhesive anchor, undercut anchor or a Hilti HSL Heavy Duty Metric Concrete Anchor.
- C. Use of expansion nail type anchors, Powder driven fasteners and friction spring - type clips are prohibited.

2.6 FABRICATION

- A. Fabricate piping and equipment supports in accordance with the requirements of paragraph 1.3 of this specification.
- B. Design hangers without disengagement of supported pipe.
- C. For hangers and supports for copper piping, provide sheet neoprene or other approved insulating material.

2.7 FINISH

- A. 316 Stainless Steel materials for "FAB" area. See Section 2.2 of this specification.
- B. Yellow dichromate conversion coat or "SUBFAB" areas. See Section 2.2 of this specification.

PART 3 - EXECUTION

3.1 GENERAL

- A. No attempt has been made to show all required pipe supports in all locations, either on the Drawings or in the details. The absence of pipe and duct supports and details on any Drawing shall not relieve the Contractor of the responsibility for furnishing and installing them as required. If 3-D modeling is used, contractor must submit 3-D hanger pipe support modeling to Engineer of Record, CM and Owner for approval.
- B. Pipe support system components shall withstand piping dead loads, thermal loads, seismic, thrust, etc. The allowable load published by the component manufacturer shall not be exceeded.
- C. All piping shall be supported in a manner that will prevent over stress on any valve, fitting, or piece of equipment. Pipe supports and hangers shall not be installed in equipment access areas, walkways or other access areas.
- D. Provide supports within 12 inch of each side of control valves 3 inch and larger.
- E. No copper pipe shall contact a pipe support or hanger of dissimilar metal. Hangers and supports for copper pipe shall be copper plated, plastic coated, or copper pipe shall be isolated with insulating neoprene strips or as approved. Tape is not an acceptable substitute for isolating copper pipe from a support or hanger of a dissimilar metal.
- F. No piping shall be supported from another pipe or any other equipment unless authorized by the Engineer.
- G. Oversized pipe hangers shall be used on all insulated piping to allow insulation to run continuous through the hanger or to allow clearance for pipe shields. Metallic pipes shall have no contact with hangers, clamps, brackets, or any other pipe support mechanism where sound and vibration control is required.
- H. Piping hanger support rods shall attach to steel beams either utilizing center-loading I-beam clamp such as Anvil Fig. 133, 134, 218, 228, or 292, or equivalent,

or welded beam attachment with a swivel connection such as Anvil Fig. 66 with Fig. 290 or 290L, or equivalent. Use of welded beam attachments without use of a swivel connection is prohibited for hot piping. C-clamp type beam clamps cannot be used on pipe sizes greater than 4 inch, unless approved by the Engineer of Record and the Owner.

- I. Install hangers without disengagement of supported pipe.
- J. Radius, deburr, and prime saw - cut ends of support components in accordance with Standard Requirements. In areas where support members, including all-thread, that encroach into aisles, egress, etc., provide protection cap on ends.
- K. Install and support nonmetallic piping in a manner to prevent adverse effects to the system due to ambient changes which occur between installation and operation.
- L. Install and support nonmetallic piping in outdoor conditions considering an exposure temperature range from minus 20 degrees F to 120 degrees F.
- M. Install rods supporting hot piping or nonmetallic outdoor piping free to pivot at the connection to the upper steel or concrete attachment.
- M. Do not allow direct contact of 16 inch and larger piping with the supporting steels such as channels, angle, or beams. Support all uninsulated piping 16 inches and larger utilizing protection saddles- Anvil Figure 160; Pipe Guides - Anvil Figure 255, 256, or Pipe Shields Inc. Model 3000 through B3300 or B 7000 through B7300, or equivalent; Pipe Slides - Anvil Figure 257, 436, 439, or Pipe Shields Inc. Model B 1000 through B 1300, or equivalent; or Pipe clamps - Anvil Figure 216 or 260.
- N. Provide variable spring hangers to accommodate expansion/contraction of pipe risers.

3.2 VIBRATION ISOLATION

- A. In accordance with Section 15200 - Vibration and Noise Control

3.3 PIPE HANGERS AND SUPPORTS

- A. Support individual horizontal piping as follows:
- B. Size hanger rod for copper pipe the same as for steel pipe.
- C. For PVC and CPVC See Section 15090 – PVC and CPVC Piping Systems.
- D. Size hanger rod for plastic pipe the same as for steel pipe. Space hangers as specified in the applicable piping specification or as shown on the piping drawings.
- E. Install hangers to provide a minimum 1/2-inch space between the face of the support and the finished covering of adjacent work with consideration for thermal movement.

- F. Support horizontal cast iron pipe adjacent to each hub with 5 feet maximum spacing between hangers.
- G. Support vertical piping at every floor, including risers in open areas. Pipe risers are not to exceed a height greater than 10 feet without adequate riser support. Strut-clamps cannot be used as a means of riser-support; only designated riser-clamp types are acceptable. Support vertical cast iron pipe at each floor at hub. Piping thermal expansion and contraction must be considered to select proper supporting components.
- H. Where several pipes can be installed in parallel and at same elevation, provide multiple or trapeze hangers.
- I. Support riser piping independently of connected horizontal piping.
- J. Do not support hot and cold piping on a common trapeze unless thermal effects are considered.
- K. For PVDF piping systems, refer to Section 15091 – PVDF Piping Systems.

3.4 PENETRATION SEALING (FIRE-RATED)

- A. Coordinate the work of this section with the requirements of Section 07841 – THROUGH-PENETRATION FIRESTOP SYSTEMS.

END OF SECTION

TABLE 1 - PIPE HANGER REQUIREMENTS FOR DEAD LOAD SUPPORTS

Pipe Size	Steel Pipe Maximum Hanger Spacing Water Service	Steel Pipe Maximum Hanger Spacing Stm. Gas or Air Service	Copper Pipe Maximum Hanger Spacing	Stainless Steel Tubing	Rod Hanger Diameter (If Applicable)
1/4 inch	5'	7'	4'	4'	3/8"
3/8 inch	5'	7'	4'	4'	3/8"
1/2 inch	5'	8'	5'	6'	3/8"
3/4 inch	7'	8'	5'	8'	3/8"
1 thru 1 ¼ inches	7'	9'	6'	8'	3/8"
1 ½ inches	9'	12'	8'	10'	3/8"
2 inches	10'	13'	8"	13'	3/8"
2 1/2 inches	11'	14'	10'	14'	½"
3 inches	12'	15'	10'	15'	½"
4 inches	14'	17'	10'	17'	5/8"
5 inches	16'	19'	10'		5/8"
6 inches	17'	21'	10'		3/4"
8 inches	19'	24'	10'		7/8"
10 inches	22'	26'			7/8"
12 inches	23'	30'			7/8"
14 inches	25'	32'			1"
16 inches	27'	35'			1"
18 inches	28'	37'			1 ¼"
20 inches	30'	39'			1 ¼"
24 inches	32'	42'			1 ½"
26 inches	32'				1 ½"
30 inches	33'				1 ¾"
36 inches	34'				2"
42 thru 60 inches	36'				2 ½"

Notes:

1. Use maximum hanger spacing unless stated otherwise on the drawings.
2. Rod hanger diameter is based upon maximum support span of water filled pipe. For shorter spans refer to hanger manufacturer's published load ratings.
3. Rods may be reduced one size for double rod hangers supporting a single pipe. Minimum rod diameter shall be 3/8 inch.
4. Hanger spacing for horizontal piping at a change of direction shall be reduced to 1/4 of the spacing specified above.

**TABLE 2 - MINIMUM DISTRIBUTION PLATE REQUIREMENTS
BASED ON 10 FOOT SUPPORT SPANS**

Pipe Size	Length/Flat Surface	Length/Pipe roller	Thickness
6 inch	3 inch	6 inch	1/4 inch
8 inch	3 inch	6 inch	1/4 inch
10 inch	6 inch	9 inch	3/8 inch
12 inch	6 inch	9 inch	3/8 inch
14 inch	6 inch	9 inch	3/8 inch
16 inch	6 inch	9 inch	1/2 inch
18 inch	6 inch	9 inch	1/2 inch
20 inch +	6 inch	9 inch	3/4 inch

Notes:

1. All approved, insulated support manufactures, regardless of model types and compressive-strength density of insert; as a minimum, shall adhere to the minimum distribution plate requirements above.
2. Hot dipped galvanized, steel load/wear distribution plates will be required for all outside, and cleanroom installations. Painted steel distribution plates are acceptable for most other interior installations.